Kuan Chieh (Louis) Lin

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EDUCATION

University of Waterloo — **Waterloo, Canada**

Bachelor of Applied Science, Major in Biomedical Engineering — *Class of 2029*

- **Academics: GPA 3.7/4.0,** Excellent Academic Standing | Scholarship: President's Scholarship of Distinction
- **Technical Skills:** SolidWorks, C++, MATLAB, Simulink, Arduino, Python, Soldering, 3D Printing, HTML, Microsoft Office
- **Certifications:** WHMIS, BioSafety, MATLAB, Simulink, TCPS 2, Machine Shop Cert

RELEVANT EXPERIENCES

Hygea Medical Technology Co., Ltd. — **Beijing, China**

Mechanical Engineering Intern — *Sept 2026 – Present*

- Developing a soft cryoablation needle for natural-orifice treatment of in situ and superficial carcinomas of the bronchial mucosa and submucosa, testing R&D prototypes to determine the strength of PEEK-to-stainless steel connections.
- Engineered Python simulations to model theoretical ice ball growth geometries within tissue phantoms, calculating surrounding power dissipation and evaluating phase-change thermodynamics.
- Conducted comparative performance testing on current-generation cryogenic needles in gelatin phantoms to evaluate freezing efficiency, internal temperature gradients, and maximum volumetric size.
- Analyzed experimental thermodynamic data to formulate heat loss equations and determine the effective treatment volume of cryogenic devices, establishing baseline metrics to guide future hardware optimizations.

Roche Canada — **Mississauga, Canada**

Pharma Technical (Quality) Intern — *Jan 2026 – May 2026*

- Engineered a custom mobile application for the PTQ Global Leaders' Summit 2026, integrating a social networking interface for executive peer-to-peer engagement; developed backend logic for real-time data synchronization, live-voting modules, and dynamic itinerary management.
- Conducted root-cause analysis on SUSVIMO implant complaints, serving as a technical supplier liaison for Phillips-Medisize to evaluate hardware compliance; performed critical failure analysis to determine the necessity of deeper investigations or regulatory recalls.
- Developed an automated data pipeline and logic-based workflow between Google Sheets and Monday.com API to eliminate manual intake latency; authored standardized operating procedures (SOPs) and video modules to facilitate technical onboarding and process scalability.

Premier Healthcare Solutions — **Taipei, Taiwan**

Medical Machinery Summer Intern — *May 2024 – August 2024*

- Monitored Varian's V-PROBE cryoprobe and provided technical support during surgeries
- Researched Varian products, including microsphere technologies, to gain a deeper understanding of their role in advancing targeted cancer therapies and modern oncology practices.
- Maintained and troubleshot Varian's V-PROBE machines through inspections, calibrations, and diagnostics, ensuring reliable performance during critical medical procedures.

UW Biomechatronics Club — **Waterloo, Canada**

Exo-Project Mechanical Team Lead — *September 2024 – Present*

- Leading the Mechanical Subteam in the design and early-stage development of a lower-body exoskeleton for the ACE 2026 Competition, drafting CAD models for a custom cycloidal gearbox and planning mechanical systems with safety and modularity in mind.
- Applied skills from previous robotic hand projects, including 3D printing, assembly, and integration of mechanical components, to develop reliable prototypes and ensure seamless hardware functionality.
- Coordinating with other subteam leads and leveraging Arduino and biosensor experience to guide system integration, iterative design, and multidisciplinary collaboration.

PROJECTS

EMG Controlled Prosthetic Hand — *May 2026 – August 2026*

- Engineered a tendon-driven, myoelectric prosthetic hand prototype using forearm sensors to translate muscle signals into mechanical actuation.
- Programmed an Arduino Uno microcontroller in C++ to process sensor inputs and drive a continuous rotation servo for dynamic grip control.
- Drafted circuit schematics and technical documentation to validate firmware and hardware integration.